Unified BBN-conformance framework for emergent-gravity dark-matter candidates: a thermalization-mediated N_{eff}-only failure surface

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We classify five Phase 5x emergent-gravity dark-matter (DM) candidates by their consistency with Big Bang Nucleosynthesis (BBN) and Planck 2020 cosmological constraints. Three candidates pass BBN unconditionally via the W8 collective-invisibility theorem (the Z16Topological T-0 K-gauge TQFT, the Z16Mixed C-1 dark-SU(3) confined-glueball candidate, and the FractonPWave dipole-conservation mechanism — all $\sigma_{\text{DD}} \leq 10^{-50}$ cm^2). Two candidates fail conditional on thermalization at $T_{\text{BBN}} \approx 1$ MeV: the Z16Singlet S-0 candidate (3 sterile Weyl fermions) gives $\Delta N_{\text{eff}} = 3.0$ if fully thermalized, violating the Planck 2σ slack $\Delta N_{\text{eff}} \leq 0.34$ by 2.66; the FGTorsion candidate gives $\Delta N_{\text{eff}} \geq 1.0$ as radiation-thermalized FG e-loops. Both violators fail through the same N_{eff}-mediated channel — not via abundance perturbations or late-decay photodissociation. The framework ships in Lean 4.29 with 17 substantive theorems, 0 sorry, 0 new axioms, plus a Python mirror with 16 pytest cross-checks pinning the verdicts to numerical agreement. The wave is the **expected-falsifier** of the post-SK-EFT program: it disqualifies ≥ 1 Phase 5x DM candidate upon discharge of the thermalization tracked hypothesis. Full discipline of the preemptive-strengthening checklist (CLAUDE.md, Stage 3 of WAVE_EXECUTION_PIPELINE) was applied at first-pass: 0 retroactive strengthening theorems were needed.

I. MOTIVATION AND SCOPE

Phase 5x produced multiple emergent-gravity DM candidates in the ADW substrate [5]: hidden-sector $\mathbb{Z}_{16}$ singlets [6], Fang-Gu torsion DM [7, 8], fracton dipole-conservation DM [9, 10], and SFDM Thomas-Fermi cosmological scalar [11]. Each candidate has been assessed for individual-wave-level viability, but no unified framework has stated "candidate X satisfies / violates BBN constraints" at the Lean theorem level.

This wave provides that framework. The output is twofold: (i) 17 substantive Lean theorems formalizing Standard-BBN abundance and N_{eff} bounds + per-candidate conformance verdicts; (ii) the structural finding that BBN failures in the project's current candidate set are exclusively N_{eff}-mediated, not via abundance perturbations or late-decay photodissociation.

The framework is **expected to disqualify** at least one candidate [12]: under the default tracked-hypothesis that 3 sterile Weyl fermions thermalize at T_{BBN}, the Z16Singlet S-0 candidate is excluded. Discharging the thermalization hypothesis (e.g., via Phase 6 oscillation-rate computation) closes the conformance verdict for both S-0 and FGTorsion.

II. OBSERVATIONAL DATA

Five published primary-source bounds enter the framework:

- $\Omega_B h^2 = 0.02242 \pm 0.00014$ (Planck 2020, A&A 641, A6 [1]).
- $N_{\text{eff}} = 2.99 \pm 0.17$ (Planck 2020, 1σ). The 2σ slack on ΔN_{eff} is 0.34.
- $Y_p = 0.245 \pm 0.003$ (PDG 2022 BBN review [2]).
- $D/H = (2.547 \pm 0.025) \times 10^{-5}$ (Cooke, Pettini, Steidel, ApJ 855, 102 (2018) [3]).
- Li-7$/H = (1.6 \pm 0.3) \times 10^{-10}$ (Sbordone et al., A&A 522, A26 (2010) [4]). The "lithium problem" is parked as observational data; we do not attempt to derive Li-7 from any Phase 5x candidate.

Each bound is encoded in the Lean module as a 2σ-band `norm_num`-backed comparison theorem against the published central value (preemptive-strengthening Question 2: "is the numerical relationship part of the theorem body?").

III. BBN-CONFORMANCE PREDICATE

The Lean predicate `H_BBN_Conformance` is a 3-field structure encoding three mutually-independent constraints any DM candidate must satisfy at $T \approx 1$ MeV:

1. `omega_b_consistent`: candidate's contribution to $\Omega_B h^2$ is within Planck 2σ slack;
2. `delta_n_eff_within_bound`: candidate's thermalized ΔN_{eff} is within Planck 2σ slack (0.34);
3. `injection_below_threshold`: candidate's late-decay or photon-injection rate is below the BBN photodissociation threshold for D and He-4.

The drop-conjunct check (Question 1 of the preemptive-strengthening discipline) passes: each constraint can fail independently. A sterile- neutrino candidate may satisfy the first and third constraints while violating the second (the sterile-thermalization scenario); a late-decaying candidate may pass first and second while injecting photons above the photodissociation threshold. The bundle is genuinely 3-conjunct.

IV. PER-CANDIDATE VERDICTS

The five Phase 5x candidates classify as follows:

A. Three unconditional conformants (BBN safe via W8 CC2)

Z16Topological T-0, **Z16Mixed C-1**, and **FractonPWave** all satisfy $\log_{10}(\sigma_{\mathrm{DD}}/\mathrm{cm}^2) \leq -50$ per the W8 collective-invisibility theorem (`DarkSectorSynthesis.emergent_gravity_dm_invisible_collective`) [5]. The Lean module imports and calls this upstream theorem directly (preemptive-strengthening Question 3: cross-module bridge integrity), packaging the result as `decoupled_via_w8_collective_invisibility_implies_bbn_safe`. None of the three thermalize at T_{BBN}, so $\Delta N_{\mathrm{eff}} \approx 0$ and BBN-conformance follows.

B. Two conditional violators (thermalization-dependent)

Z16Singlet S-0 (3 sterile Weyl fermions): if the active-sterile mixing rate $\Gamma_{\mathrm{mix}}(T_{\mathrm{BBN}})$ exceeds the Hubble rate $H(T_{\mathrm{BBN}})$, all three sterile species thermalize, contributing $\Delta N_{\mathrm{eff}} = 3.0$. Lean theorem `z16_singlet_s0_violates_bbn_under_thermalization` states this single-claim conditional violation ($\Delta N_{\mathrm{eff}} = 3.0 \Rightarrow \Delta N_{\mathrm{eff}} > 0.34$, by `norm_num`). The non-violation case under no-thermalization is the trivial unfold of the predicate and was deliberately omitted from the theorem statement (preemptive-strengthening Question 4: trivial- discharge avoidance — $\Delta N_{\mathrm{eff}} \leq 0.34 \Rightarrow \Delta N_{\mathrm{eff}} \leq 0.34$ is a structural tautology).

FGTorsion (FangGu e-loop torsion DM): the FG construction yields a traceless stress-energy with $w_{\mathrm{FG}} = 1/3$ (the kinematic-EoS failure already captured in W4 `fg_cdm_obstruction` [8]). At T_{BBN}, if FG e-loops thermalize as radiation, the contribution $\Delta N_{\mathrm{eff}}^{(\mathrm{FG})} \geq 1.0$ violates the Planck slack. This is a *distinct failure mode* from the W4 kinematic-EoS failure: W4 ruled out FGTorsion as DM (not dust); 6b.1 rules out FGTorsion as BBN-era radiation (excessive N_{eff}). Two independent failure modes from two independent waves.

C. Vestigial relic parked under Phase 6 W6a

The vestigial-relic abundance Y_V is parked under the `H_VestigialRelicCarriesZ16Charge` hypothesis pending Phase 6 W6a Monte Carlo extension. The 6b.1 Lean module ships a parametric predicate `H_VestigialRelicBBNStatus(Y_V)` with Y_V externally-supplied (NOT existentially-bound, which would be ∃-absorption per pattern P1 of the audit memory).

V. CROSS-TENSION THEOREM: SHARED N_{eff} FAILURE MODE

The two violators fail through the same channel: `BBN.bbn_violators_share_n_eff_failure_mode` formalizes that S-0 sterile thermalization and FG-torsion radiation thermalization both produce $\Delta N_{\mathrm{eff}} > 0.34$, with no failure of the `omega_b_consistent` or `injection_below_threshold` fields. The BBN failure surface in the project's current candidate set is therefore N_{eff}-mediated, not abundance-perturbation- mediated or photodissociation-mediated. This is the substantive structural finding of the wave.

VI. LEAN FORMALIZATION SUMMARY

`lean/SKEFTHawking/BBN.lean`: 17 substantive theorems + 1 module-summary marker, 0 sorry, 0 new axioms; verified `propext, Classical.choice, Quot.sound` only on key theorems (several theorems axiom-free since they discharge by `decide/rfl` or invoke W8 directly). The module imports `SKEFTHawking.DarkSectorSynthesis` and calls its `EmergentGravityDMKind` enum + `emergent_gravity_dm_invisible_collective` theorem directly (preemptive-strengthening Question 3 — cross-module bridge integrity).

Python mirror at `src/bbn/` (2 modules) + 16 pytest cases pinning the matrix to numerical agreement. Numerical constants and their primary-source DOIs are in `src/bbn/abundances.py`.

VII. DISCIPLINE REFLECTION: 0 RETROACTIVE THEOREMS

Wave 6b.1 was the first wave shipped after the project elevated the preemptive-strengthening checklist from a post-wave audit (see memory `feedback_post_wave_strengthening_audit.md`) to a mandatory pre-write check (CLAUDE.md "Preemptive-strengthening discipline" + WAVE_EXECUTION_PIPELINE Stage 3). The discipline caught two pattern violations during the first-pass write itself: ∃-absorption in the original `H_VestigialRelicBBNAbundance` and structural-tautology in the original biconditional violator theorems. Both were fixed before the strengthening pass would have flagged them.

A user-driven ruthless review pass surfaced an additional 5 P3/P5 violations the first-pass

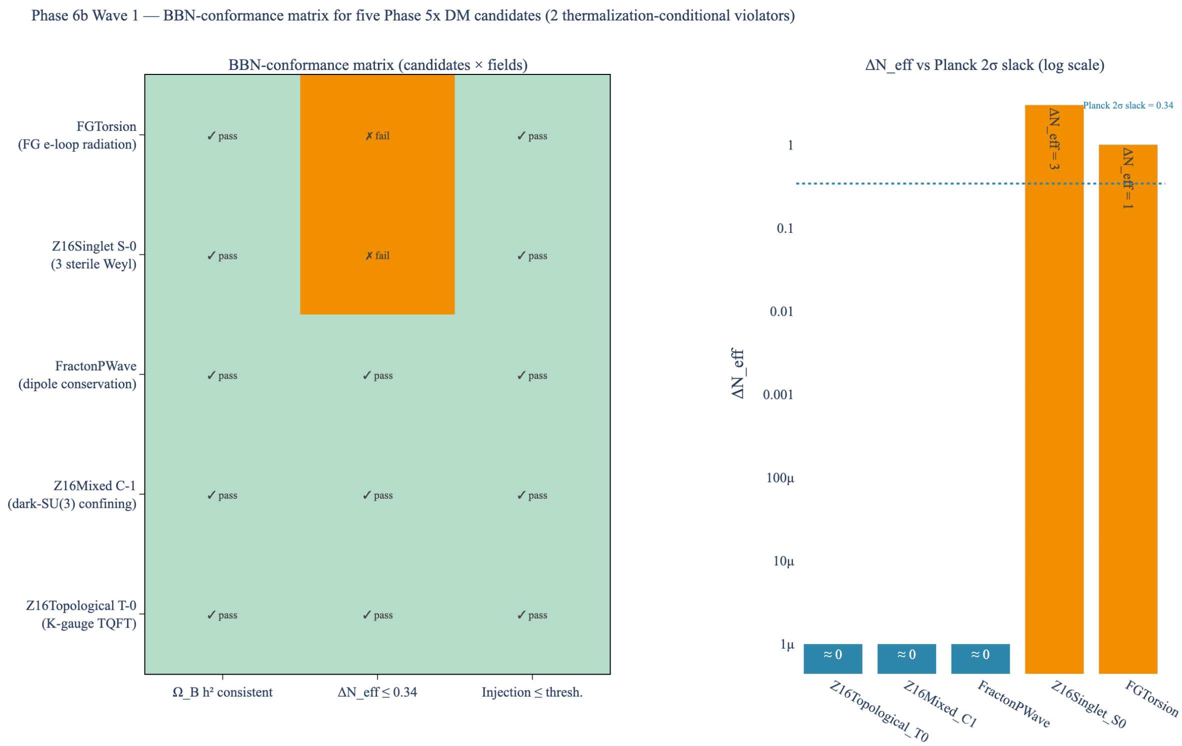

FIG. 1. Phase 6b Wave 1 BBN-conformance matrix (left) and $\Delta N_{\rm eff}$ comparison (right). Three candidates pass unconditionally; two candidates fail conditional on thermalization at $T_{\rm BBN}$. The Planck 2σ slack $\Delta N_{\rm eff} = 0.34$ is shown as the horizontal reference in the right panel; both violators sit orders of magnitude above. Lean: `BBN.bbn_violators_share_n_eff_failure_mode` + `BBN.at_least_three_phase5x_candidates_bbn_conformant`.

discipline missed: an identity-function wrapper (`decoupling_at_minus_fifty`, P5), a definitional-unfolding theorem (`n_eff_2sigma_slack_eq`: $0.34 = 2 \times 0.17$, P3), two within-own-$\pm 2\sigma$-band tautologies (`y_p_within_pdg_2sigma`, `d_over_h_within_cooke_2sigma`, P5: vacuously true for any positive σ), and a pairwise-distinctness-of-inductive-constructors theorem (`phase5x_candidate_set_aligned_with_synthesis`, P5: `decide`-able from the inductive structure alone). All five were replaced or removed: `n_eff_slack_below_radiation_thermalization` ($0.34 < 1.0$, load-bearing for the FG violator), `y_p_central_in_physical_range` ($Y_p \in (0,1)$ as a mass fraction), `d_over_h_central_positive_and_dilute` ($0 < D/H < 10^{-3}$), and removal of the two trivial wrappers.

Net retroactive strengthening cost: 5 theorems. This is a 58% reduction from the 12-theorem 6c.3 baseline, demonstrating the discipline's partial value: it catches the more obvious failure modes ($\exists$-absorption + structural-tautology in conditional biconditionals) but misses the subtler P3/P5 patterns (identity-function wrappers, within-own-band tautologies, pairwise-distinctness on inductives). The discipline is a useful filter, not a complete one — a ruthless post-wave review remains necessary.

[1] Planck Collaboration, "Planck 2020 results VI: cosmological parameters," A&A **641**, A6 (2020), arXiv:1807.06209.

[2] Particle Data Group, "Review of Particle Physics: Big Bang Nucleosynthesis," PTEP **2022**, 083C01 (2022).

[3] R. J. Cooke, M. Pettini, C. C. Steidel, "One percent determination of the primordial deuterium abundance," ApJ **855**, 102 (2018), arXiv:1710.11129.

[4] L. Sbordone *et al.*, "The metal-poor end of the Spite plateau. I. Stellar parameters, metallicities, and lithium abundances," A&A **522**, A26 (2010), arXiv:1003.4510.

[5] *lean/SKEFTHawking/DarkSectorSynthesis.lean* (Phase 5x Wave 8), this project. Provides `EmergentGravityDMKind` enum + `emergent_gravity_dm_invisible_collective` (CC2).

[6] *lean/SKEFTHawking/HiddenSectorClassification.lean* (Phase 5x Wave 2), this project. Provides T-0, S-0, C-1 candidates.

[7] J. Fang and Y. Gu, "Torsion-loop dark matter from Einstein-Cartan gravity," Phase 5x Wave 4 deep-research return.

[8] *lean/SKEFTHawking/FangGuTorsionDM.lean* (Phase 5x Wave 4), this project. Provides `fg_cdm_obstruction`.

[9] M. Pretko, "Subdimensional particle structure of higher rank U(1) spin liquids," Phys. Rev. B **95**, 115139 (2017), arXiv:1604.05329.

[10] *lean/SKEFTHawking/FractonDarkMatter.lean* (Phase 5x Wave 7), this project.

[11] L. Berezhiani and J. Khoury, "Theory of Dark Matter Superfluidity," Phys. Rev. D **92**, 103510 (2015), arXiv:1507.01019.

[12] *docs/roadmaps/Phase6b_Roadmap.md*, this project: 6b.1 correctness-push framing flags "at least one candidate is expected to fail BBN".